SMART SPEND: MACHINE LEARNING-POWERED PERSONAL MONTHLY EXPENSE PREDICTOR

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ABSTRACT

Predicting personal expenses can support financial planning, but spending is irregular and may be difficult to estimate from memory alone. This study documents the design and demonstration of Smart Spend, a Python-based expense tracking and prediction system that integrates transaction recording, data preprocessing, exploratory analytics, and supervised regression. The group-prepared dataset contains 528 transactions across 24 complete months, including 480 expense and 48 income transactions. Expense records were aggregated into monthly totals and converted into lagged, rolling, trend, and seasonal features. Ridge Regression, Random Forest, and a last-period baseline were compared using a chronological test set. In a reproducible evaluation of the included dataset, Random Forest obtained the lowest mean absolute error (PHP 887.25), compared with Ridge Regression (PHP 1,129.38) and the baseline (PHP 1,268.52). This finding only describes performance on the project dataset and does not establish accuracy for real users. Smart Spend also offers early, explicitly unvalidated estimates when users have only three complete months or six complete weeks of records. The system demonstrates an end-to-end learning and visualization workflow while identifying privacy, dataset representativeness, and deployment persistence as major limitations.

Key terms: Personal finance; Expense prediction; Machine learning; Ridge Regression; Random Forest; Time-series evaluation.

I. INTRODUCTION — CHAPTER 1

Background of the Study

Financial awareness often depends on consistent recording of purchases, transportation costs, utility payments, and other expenses. Manual budgeting can show what has already been spent, but anticipating a future month requires interpreting patterns in previous periods. Smart Spend addresses this need through a single interface that records expenses, summarizes spending, and produces a forecast from historical data. Its outputs are intended to support reflection and planning, not to guarantee future financial behavior.

Problem Statement

The project addresses three related problems: (1) fragmented or incomplete personal expense records; (2) limited understanding of spending distributions and changing totals; and (3) difficulty estimating expenses for an upcoming month or week. A useful system must also distinguish a simple estimate based on short history from a trained and evaluated machine-learning result.

Objectives

The general objective is to develop and evaluate a Python-based application that transforms transaction data into summaries and expense forecasts. Specifically, the study aims to: collect and validate user records or CSV datasets; calculate descriptive statistics and at least three visualizations; prepare historical spending features; compare regression methods and a baseline using appropriate error metrics; provide interpretable forecasts; and document limitations and privacy considerations.

Scope and Significance

The implementation covers account registration, login, expense and income entries, user-selected monthly budgets, manual weekly/monthly history totals or category breakdowns, CSV import, analytics, and a read-only administrator view. The study evaluates the dataset prepared for this project; it does not claim a population-representative sample. A single-user account's forecasts depend on its own available historical transactions. The system offers a practical example of how applied data science can complement expense monitoring.

II. RELATED SYSTEMS AND CONCEPTS — CHAPTER 2

Regression-Based Forecasting

Regression predicts a numerical target using features derived from historical observations. Ridge Regression applies L2 regularization to linear regression, which can reduce unstable coefficients when predictors are correlated (Hoerl & Kennard, 1970). Random Forest combines decision trees and can represent nonlinear relationships (Breiman, 2001). Smart Spend implements both algorithms as alternatives rather than assuming either is always superior.

Time-Aware Evaluation and Responsible AI

For sequential observations, evaluating a model on later periods is more appropriate than randomly mixing past and future periods. The scikit-learn documentation explains that time-ordered splitting reduces the risk of assessing performance with future information (scikit-learn developers, n.d.-a). Error measures such as MAE, RMSE, and R^2 reveal different aspects of prediction quality

(scikit-learn developers, n.d.-b). The NIST AI Risk Management Framework emphasizes validity, reliability, transparency, and privacy considerations for AI-enabled systems (NIST, 2023).

III. METHODOLOGY — CHAPTER 3

Research and Development Design

The project follows a developmental and comparative experimental approach. The website and API were implemented in Python with a SQLite database, while the forecasting functions use NumPy and scikit-learn. The group prepared a reproducible project dataset to demonstrate ingestion, preprocessing, feature engineering, evaluation, and visualization. The records are project-created examples, not data collected from study participants.

Data Source and Description

The repository includes sample_data/demo_transactions.csv and a data dictionary. The group created the sample transaction records for project testing; they were not collected from real individuals. A total of 528 records span November 2024 to October 2026: 480 expenses and 48 income entries. The transaction categories include Food & Drinks, Transportation, Shopping, Bills & Utilities, Health, Entertainment, Education, and Income. Records are organized by date and category.

Table 1. Dataset fields and their roles

Field	Description
date	ISO date (YYYY-MM-DD)
description	Transaction label
category	Expense category or Income
amount	Positive PHP value
type	expense or income

Data Collection and Validation

Data may enter the system through individual transactions, completed month/week summaries, category breakdowns, or CSV upload. The import routine validates dates and positive amounts, standardizes recognized category aliases, skips invalid records, and rejects exact duplicates. Transactions above PHP 100,000 are flagged for review rather than silently removed. Monthly and weekly summary entries cannot overlap with detailed expense records for the same covered interval; the application blocks such overlaps to avoid double counting.

Preprocessing and Feature Selection

Only expense transactions contribute to the forecasting target. Monthly totals are obtained by grouping dates into YYYY-MM buckets; weekly totals use Monday–Sunday periods. Incomplete current periods and missing periods are excluded rather than interpreted as zeros. The predictive features comprise the previous three period totals, their three-period average, a recent trend term, and two seasonal sine/cosine components. Features are calculated solely from periods earlier than the predicted target.

Table 2. Preprocessing decisions

Issue	Applied handling
Invalid date / amount	Skip invalid CSV row
Category variations	Normalize to known categories
Duplicate record	Skip duplicate
Large amount	Flag > PHP 100,000
Incomplete periods	Exclude from training
Missing period	Do not impute zero

Model Training and Experimentation

When there are at least 12 consecutive completed months (or 14 consecutive weeks), the system builds supervised examples from lagged history. For the monthly demonstration, 24 months produce 21 feature/target pairs; the final six examples form a chronological holdout, leaving 15 examples for training. Ridge Regression uses standardized features with alpha = 15. Random Forest uses 120 trees, maximum depth 5, minimum leaf size 2, and random_state = 42. A last-period-value baseline is also tested. The model with the lowest holdout MAE is selected; if it is an ML estimator, it is refitted on all available completed periods before forecasting.

With only 3–11 complete months or 6–13 complete weeks, Smart Spend provides a recency-weighted estimate with a capped trend adjustment. The interface explicitly labels such a result as an early estimate: it is not presented as trained machine learning and is not assigned an accuracy score.

Evaluation Measures

MAE averages the absolute differences between actual and predicted period totals. RMSE takes the square root of mean squared error and penalizes larger mistakes more strongly. R^2 summarizes model fit relative to a mean-only reference; a negative score is possible. Lower MAE and RMSE are desirable, but even favorable holdout values cannot establish reliability outside the data represented by the experiment.

Software Architecture and Ethics

The workflow is: user/CSV entry → validation and SQLite storage → grouping and feature preparation → regression/baseline comparison → forecast and dashboard charts. Sessions and account roles are handled by the Python server; passwords are stored as salted hashes. Admin functions can inspect user records as part of the demonstration role. The public test deployment currently uses ephemeral SQLite storage on Render Free, which may lose records after a restart; real financial information should not be entrusted to that configuration.

IV. RESULTS AND DISCUSSION — CHAPTER 4

Dataset Exploration

The group-prepared dataset comprises 24 complete months of records, with 480 expense transactions and 48 income transactions. Total recorded expenses are PHP 302,884.50. Mean monthly expense is PHP 12,620.19, with observed monthly totals ranging from PHP 8,872.84 to PHP 15,250.67. These figures describe this project

dataset and should not be interpreted as Philippine household averages.

The figures below are screenshots of the live Smart Spend Analytics page rather than separately generated charts. Figure 1 shows the latest 12-month spending trend; Figures 2 and 3 summarize the October 2026 expenses. The category bars in Figure 3 total PHP 27,707.26, with Bills & Utilities (PHP 10,253.18) and Food & Drinks (PHP 9,166.06) as the two largest categories. The interface rounds the composition percentages, so displayed percentages may not sum to exactly 100%. These views reflect the records present in the illustrated website account and should not be interpreted as independently evaluated forecast results.

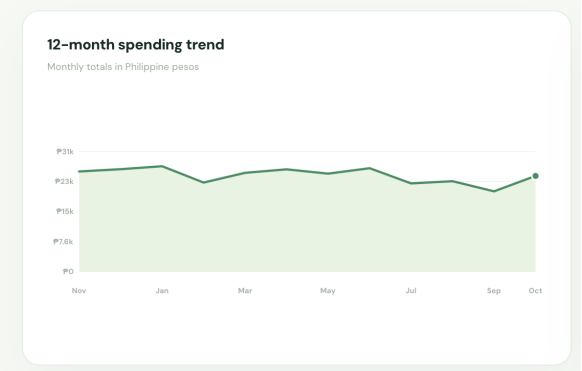


Figure 1. Smart Spend website: 12-month spending trend (November 2025–October 2026).

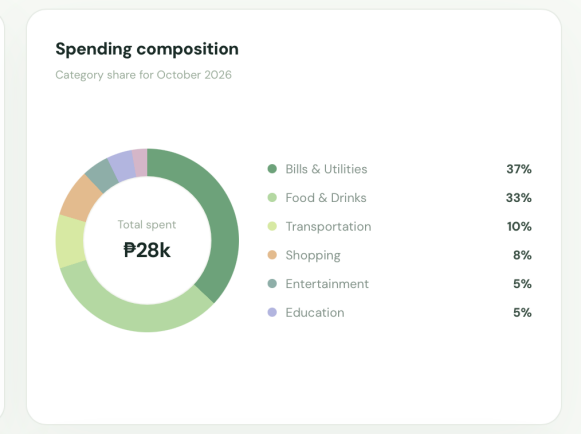


Figure 2. Smart Spend website: spending composition for October 2026.

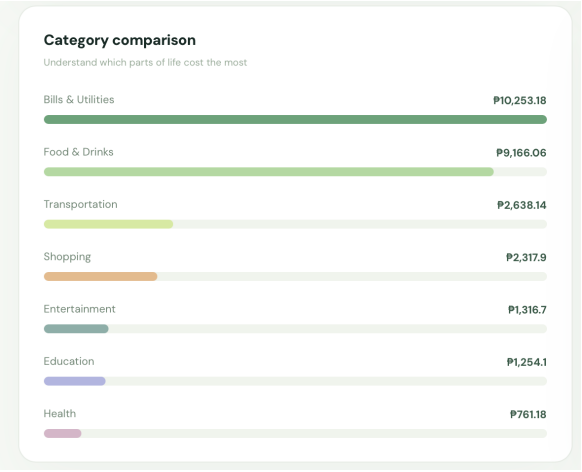


Figure 3. Smart Spend website: October 2026 expense-category comparison.

Model Comparison Results

A reproducible run of the project's fit_and_forecast function was performed on November 9, 2026, using the sample dataset and monthly forecasting mode. All reported metrics below are calculated by the actual included program, not estimated manually. The holdout comprises the final six monthly prediction samples; model selection is based on the lowest MAE.

Table 3. Measured chronological holdout results

Model	MAE	RMSE	R^2
Ridge Regression	1,129.38	1,216.31	-0.279
Random Forest	887.25	998.19	0.139
Last-period baseline	1,268.52	1,475.19	-0.881

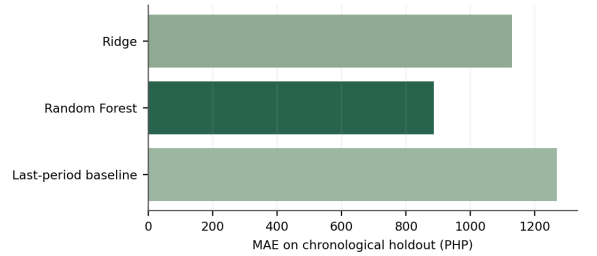


Figure 4. MAE comparison; lower values indicate smaller average errors.

Random Forest achieved the lowest holdout MAE (PHP 887.25) and RMSE (PHP 998.19), with $R^2 = 0.139$. Ridge Regression recorded MAE = PHP 1,129.38 and $R^2 = -0.279$, while the last-period baseline recorded MAE = PHP 1,268.52 and $R^2 = -0.881$. Although Random Forest performed best among the three on this single holdout on the project data, the small test size and low R^2 indicate limited explanatory strength. Additional representative records and repeated prospective tests would be necessary before making real-world accuracy claims.

Forecast Example and Interpretation

Using data completed through October 2026, the module projected the incomplete November 2026 month at PHP 13,567.64 and predicted December 2026 expenses of PHP 14,147.57. This forecast is based on the project dataset and is not a verified future outcome. It illustrates the pipeline's ability to produce a period-specific result and attach a selected model plus evaluation notes.

Table 4. Project experiment summary

Measurement	Observed program result
Expense history	24 complete months
Train / holdout pairs	15 / 6
Selected model	Random Forest
Next-month example	PHP 14,147.57 (Dec 2026)
Real-world validation	Not established

Discussion of Requirements and Limitations

The functional system implements dataset loading, validation, descriptive statistics, multiple charts, supervised forecasting, measured model comparisons, and readable summaries. The regression experiment meets the requirement to compare methods; however, the project-created records are not an independent sample of real

households. Their patterns may differ from actual household spending. Changes in employment, emergencies, inflation, missing transactions, and irregular purchases can invalidate predictions. An administrator's ability to inspect financial records is also a privacy concern that requires access control, retention rules, and informed users before actual deployment.

V. CONCLUSION AND RECOMMENDATIONS — CHAPTER 5

Smart Spend demonstrates a complete small-scale data science application for transaction analysis and monthly or weekly expense forecasting. The Python system collects and cleans data, displays descriptive summaries and visualizations, and implements a model comparison using a time-ordered holdout. In the 24-month project evaluation, Random Forest produced the lowest MAE among the evaluated alternatives. These results support the technical feasibility of the workflow, but they do not establish real-world predictive accuracy or the suitability of the current Render Free database for personal financial information.

Future work should include consented and anonymized real-world records across diverse spending patterns, prospective forecast tracking, stronger baseline and seasonal comparisons, repeated rolling-origin evaluation, and clearer explanation of uncertainty. A durable hosted database, stronger operational security, role-specific privacy controls, and longer-term usability testing are recommended before public financial use.

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APPENDIX A. REPRODUCIBILITY AND FILES

Public source code: https://github.com/dandan1022/SMART-SPEND . Live classroom demonstration: https://smart-spend-ql5m.onrender.com . Dataset: sample_data/demo_transactions.csv . Dataset dictionary: sample_data/DATA_DICTIONARY.md . ML implementation: ml_model.py . To reproduce the reported monthly model evaluation locally, use the included sample expense rows as date and amount_cents inputs to fit_and_forecast, with today = date(2026, 11, 9) and period = "monthly". The result depends on these fixed inputs and on compatible installed library versions.

APPENDIX B. SYSTEM TEST AND USER GUIDE

Demonstration sequence: (1) create a user account; (2) enter an expense or import the provided project CSV; (3) review Dashboard and Analytics charts; (4) open AI Predictor and compare models; (5) inspect data validation and model notes; and (6) sign in using the separately configured administrator account to review the read-only database dashboard. The repository includes automated tests for authentication, history entry, and category breakdown, but this paper does not claim that all deployment scenarios have been independently verified.

APPENDIX C. PRESENTATION EVIDENCE

Recommended screenshots for the final submission: registration and login, dashboard, dataset import and cleaning summary, three analytics visualizations, trained model evaluation table, predicted expense output, and administrator view. Insert current screenshots captured from the actual deployed website before submission if your instructor requires image evidence.